

Lab Assignment 1

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1 Task 1

The aim of *Task 1* was to calculate the first three resonance frequencies of the beam under Free-Free conditions in y-direction using FE-software or analytical expressions. This was done in two ways - in Matlab using Calfem and in Abaqus CAE.

The two methods gave slightly different results, it is likely that this is because the two methods solves two different beam models, that is Euler-Bernoulli (Matlab) and Timoshenko (Abaqus).

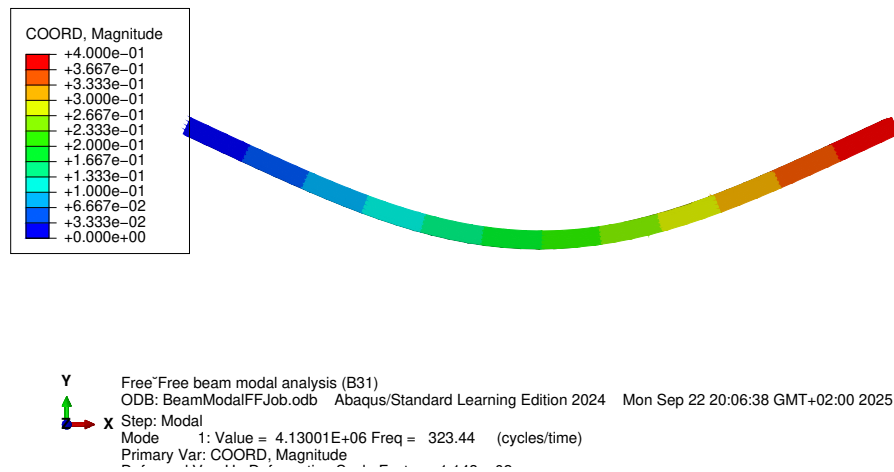


Figure 1: Abaqus first Mode shape and resonant frequency

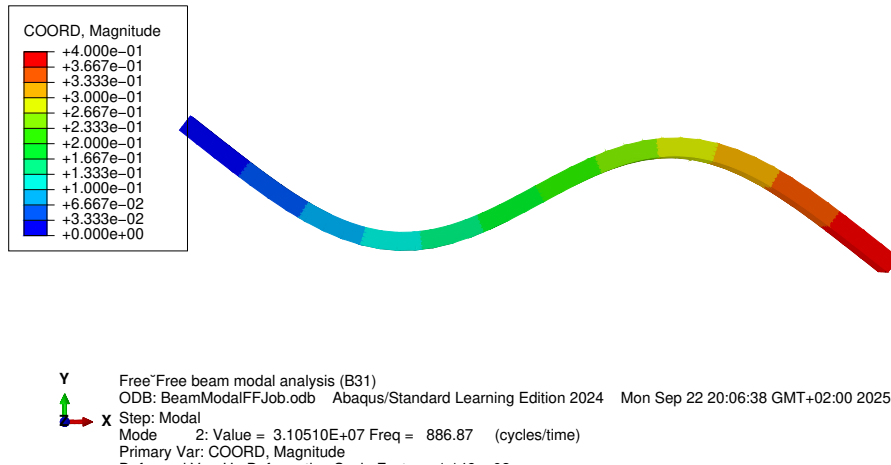


Figure 2: Abaqus second Mode shape and resonant frequency

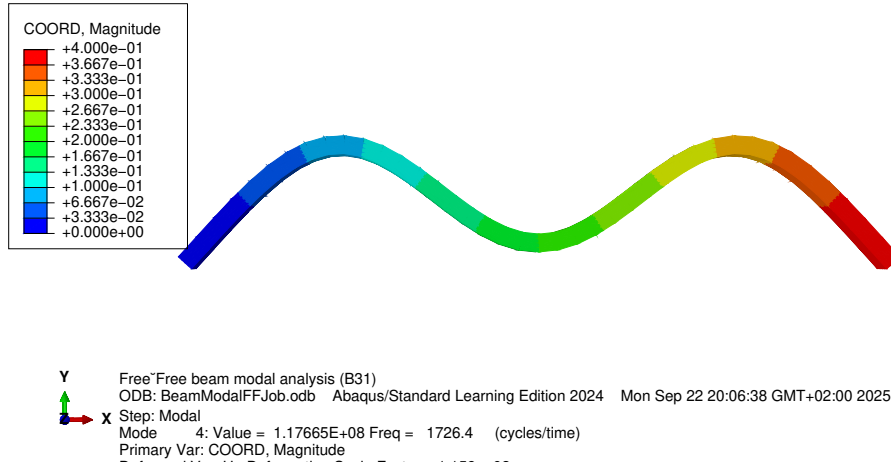


Figure 3: Abaqus third Mode shape and resonant frequency

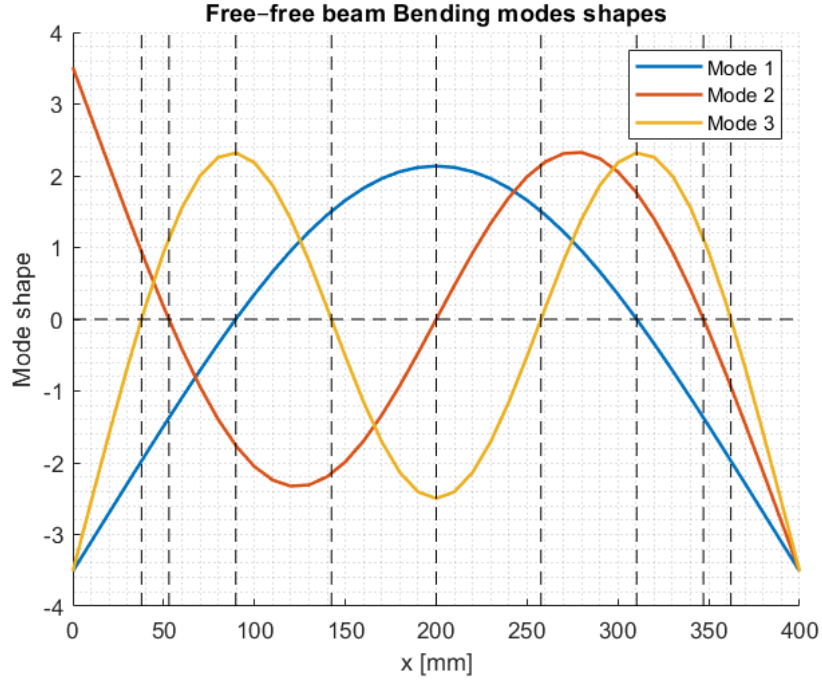


Figure 4: Matlab plot of first three mode shapes for Free-Free Beam.

The black vertical dotted lines in [fig.4] show where any of the three modes are always zero, and shows where not to do measurements.

Mode	Abaqus f [Hz]	Matlab f [Hz]
1	323.44	324.77
2	886.87	895.25
3	1726.40	1755.06

Table 1: Resonant frequencies for first three modes - Abaqus vs Matlab.

2 Task 2

The aim of *Task 2* was to calculate the first three resonance frequencies of the beam under Fixed-Free conditions (cantilever) in y-direction using FE-software or analytical expressions. This was done in Matlab only, using Calfem Euler-Bernoulli model.

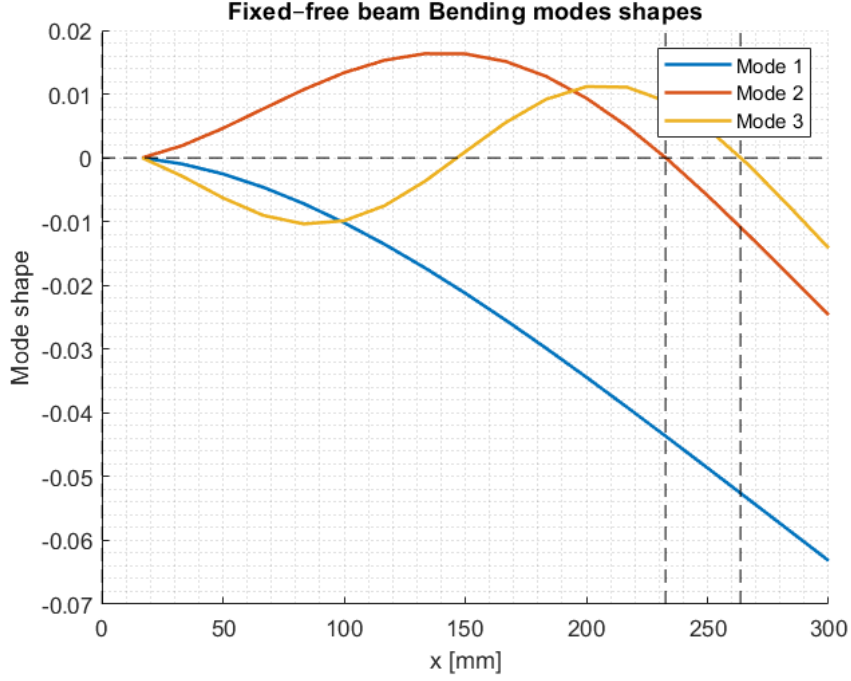


Figure 5: Matlab plot of first three mode shapes for Fixed-Free Beam.

The black vertical dotted lines in [fig.5] show where any of the three modes are always zero, and shows where not to do measurements.

Mode	Matlab f [Hz]
1	90.74
2	564.64
3	1592.23

Table 2: Resonant frequencies for first three modes - Abaqus vs Matlab.

3 Task 3

The highest frequency in Task 1 and in Task 2 is 1755 Hz, rounding up to 2000 Hz so we set the sampling frequency to $f_s = 4000$ Hz.

4 Task 4

The experimental data of the Free-Free beam is given by the last column in [tab.3]

Mode	Abaqus f [Hz]	Matlab f [Hz]	Experimental f [Hz]
1	323.44	324.77	326.17
2	886.87	895.25	891.11
3	1726.40	1755.06	1736.82

Table 3: Resonant frequencies for first three modes - Abaqus vs Matlab vs Experimental Data.

As seen in [tab.3], the Euler-Bernoulli based model that is used in Matlab follows the experimental values better than the results from Abaqus for the lower frequencies, while Abaqus is more coherent with the experiment for the highest mode that is observed. The experiment was conducted with five averages. It was decided to both excite and measure the acceleration at the end of the beam, based on the results from [fig.4]

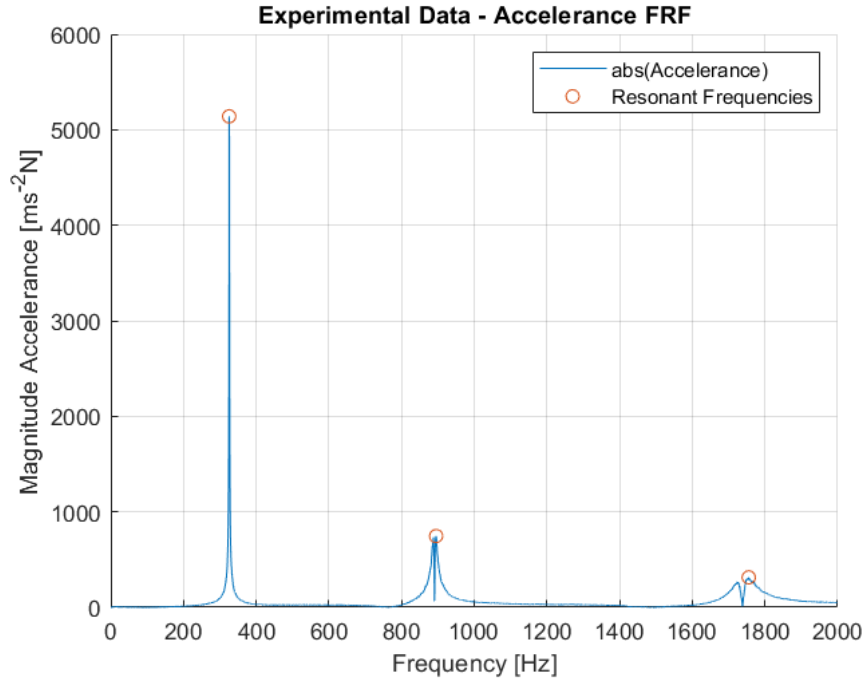


Figure 6: Magnitude of accelerance FRF from experimental Free-Free beam.

5 Task 5

For the experiment of a cantilever beam, we decided to excite the same point every time, point 5. We took 5 samples of each measuring point, averaged it and moved the accelerometer from point 1 (free end) to 6 (one from fixed end). For each point we got one FRF, these are given in [fig.7].

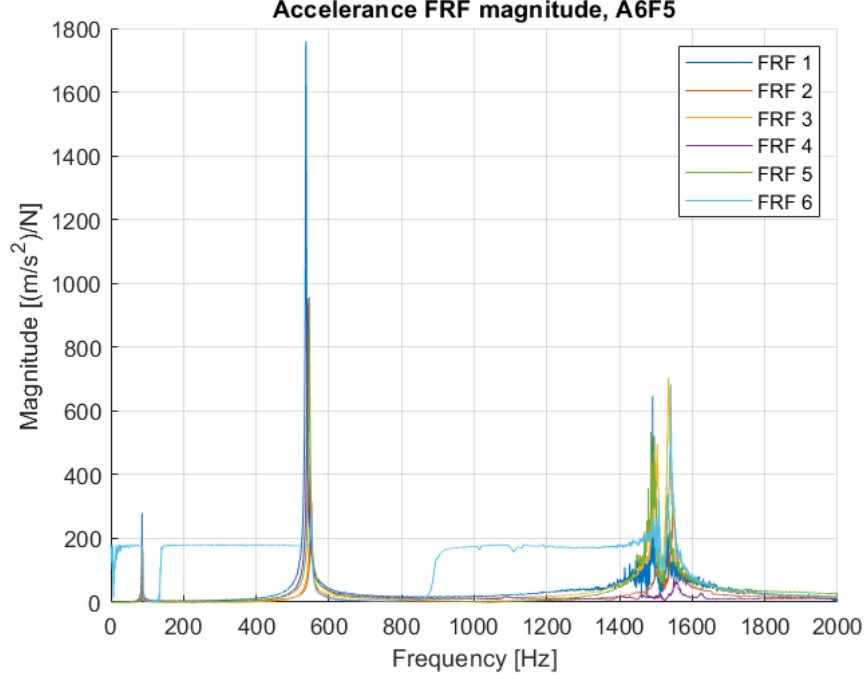


Figure 7: All FRF's (Accelerance) in one plot.

As seen, the first FRF has some issues but overall, the resonant frequencies are not shifted so it is useful information still.

6 Task 6

Comparing the resonance frequencies from the experiment in Task 4 with the simulated resonance frequencies from the free-free beam, we can see that it's not quite right. The given Youngs modulus of $E = E_0 = 69$ GPa is updated (scaled) by

$$scale_E = (f_{simulation}^{-1} f_{experiment})^2,$$

since

$$K = \omega^2 M \rightarrow (M^{-1}K)_{sim} = \omega_{sim}^2$$

$$\frac{(M^{-1}K)_{exp}}{(M^{-1}K)_{sim}} = \left(\frac{\omega_{exp}}{\omega}\right)^2 = \left(\frac{f_{exp}}{f_{sim}}\right)^2$$

So the new Youngs modulus is given by $E_{new} = \left(\frac{f_{exp}}{f_{sim}}\right)^2 E_0 = 67.78$ GPa.

Mode	Experiment f [Hz]	Difference with $E_0 = 69GPa$ [Hz]	Difference with $E_{new} = 67.78GPa$ [Hz]
1	326.17	-1.40	-4.27
2	891.11	+4.14	-3.76
3	1736.82	+18.25	+2.87

Table 4: Resonant frequencies for different modes with different Youngs modulus.

As seen in [tab.4], the errors does not disappear, since we only scale the modulus by a scalar and the errors between the first simulation and the experiment are both positive and negative. We have however 'normalized' the errors, we no longer have the larger error of 18.25 Hz.

7 Task 7

Plot the coherence and FRF from one of the measurement points. One of the best measurements we did was for response in point 2, it shows in [fig.8], where the coherence were over 80% for 98.7% of all datapoints.

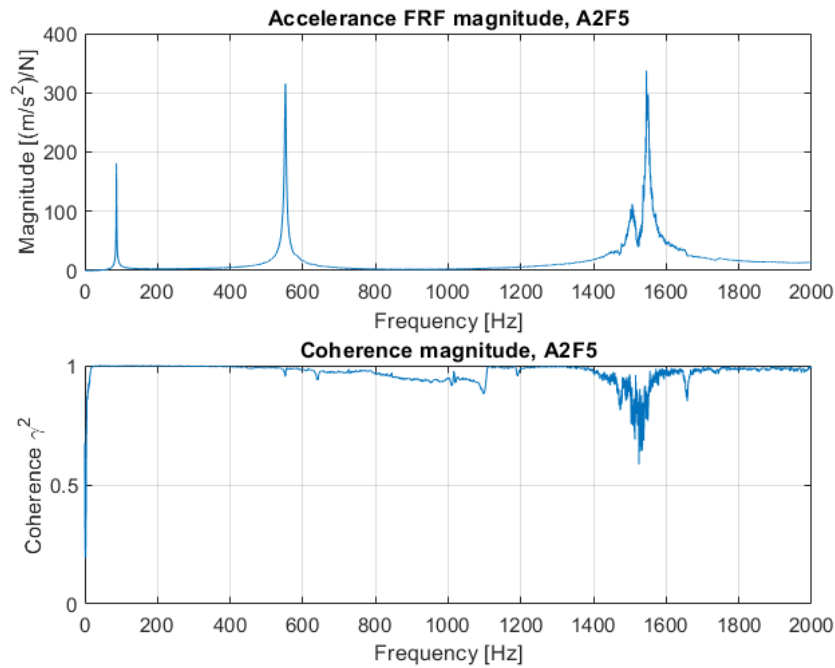


Figure 8: Coherence of measurement with response at point 2, reference at point 5.

We can see in the stability plot four peaks. Based on the simulation, the last two peaks are assumed to be the same resonance frequency.

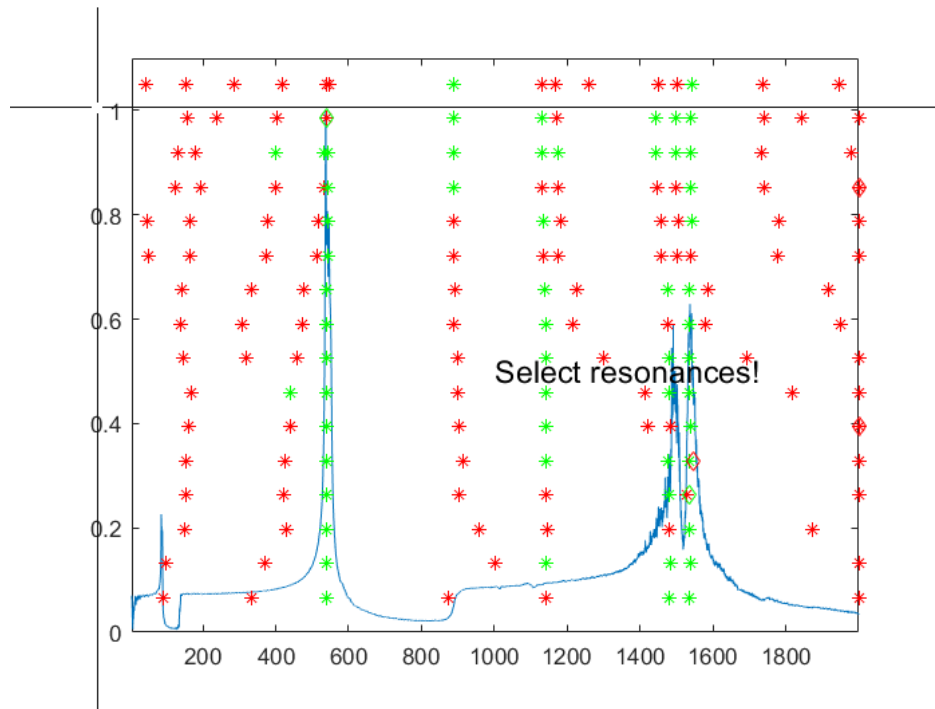


Figure 9: Stability Plot

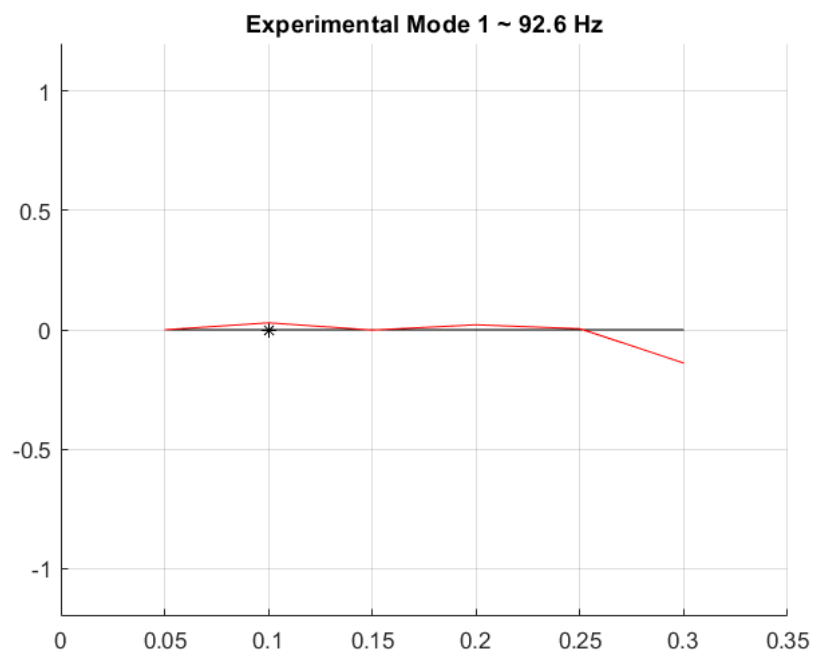


Figure 10: Mode 1 Experimental



Figure 11: Mode 2 Experimental

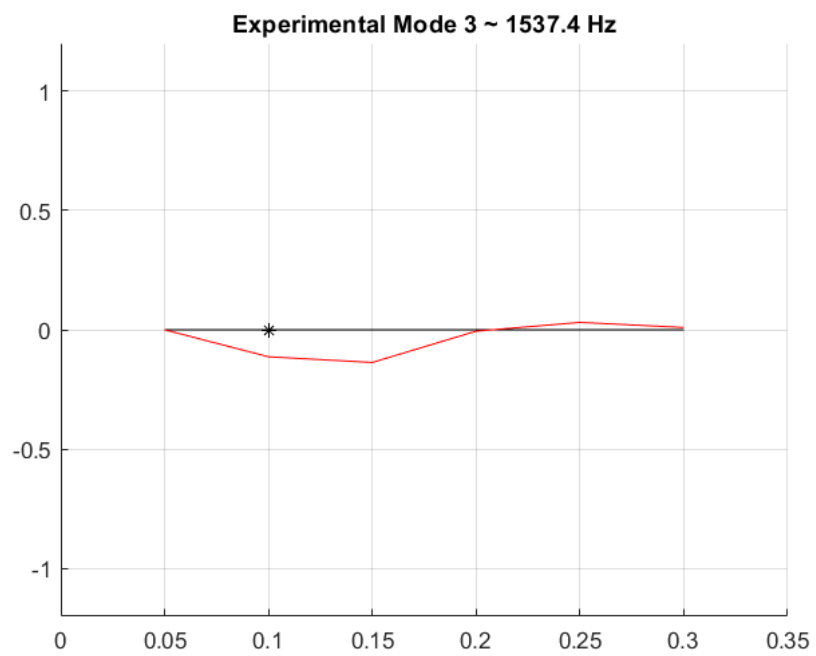


Figure 12: Mode 3 Experimental

8 Task 8

By instead of just forcing an ideally clamped fixed end, introducing large springs that hold the cantilever beam end 'kind of' fixed, and adding a smaller but still large spring to keep the slope at the fixed end, the resonant frequencies fit much better. This was done by trial and error, and the results are visible in [fig.13] where the first bar of each mode-group represents the experimentally measured resonant frequency, the middle bar the spring-based fixed end and the last bar the ideally fixed end but with updated Youngs modulus. It is shown that with $k_\theta = 50 \text{ kNm/rad}$, the overall resonant frequencies match the experiment better.

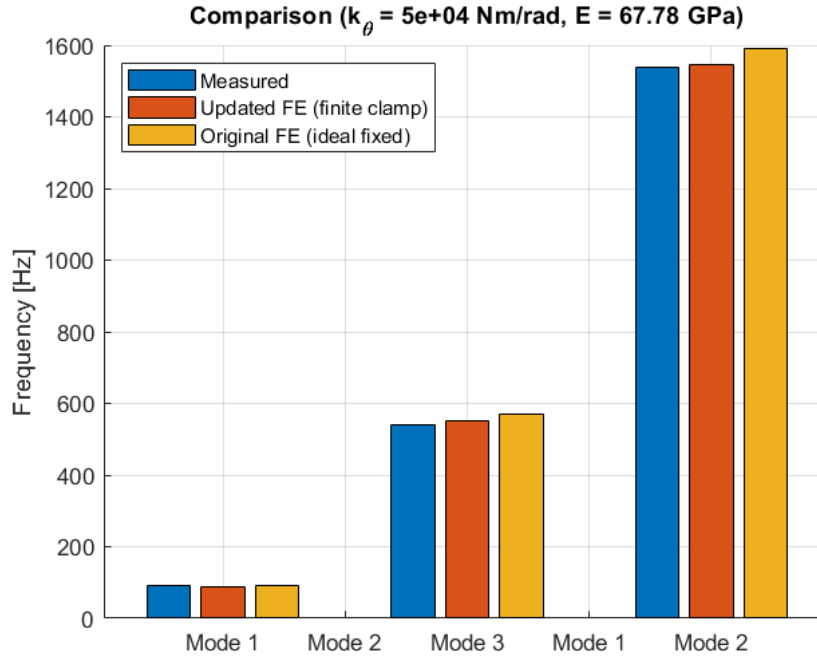


Figure 13: Optimized clamping conditions.

8.1 Appendix A, MATLAB Code

Listing 1: MATLAB script for Lab 1

```
%% Task 1 - Free-Free Sim
clear all; close all; clc;
L = 400e-3; % [m]
Nel = 40; % Number of elements
modes = [1 2 3];
E0 = 69e9;

[M, K] = calfem_beam(Nel, L, E0);

C = 0*M; % Undamped
inno = 1; % Input DOF (y at node 1)
y_pos = 2:3:3*(Nel+1); % outno DOFs: (ux, uy, theta)
npoles = 3;

[poles, residues] = resonances(M, C, K, inno, y_pos, npoles);
[fn, zeta] = poles2fd(poles);
```

```

% Solve eigenproblem  $K\psi = \omega^2 M\psi$ 
[psi, D] = eig(K, M);          %  $D = \text{diag}(\omega^2)$ 
w = sqrt(diag(D));
f_freefree_sim = w/(2*pi); % Natural frequencies [Hz]

% Sort by frequency
[f_freefree_sim, idx] = sort(f_freefree_sim);
psi = psi(:, idx);
psi = psi(:, f_freefree_sim > 0);

f_freefree_sim = f_freefree_sim(f_freefree_sim > 0);
f_freefree_sim = f_freefree_sim(1:3);

x = linspace(0, L*1e3, Nel+1);          % x in mm for plotting

figure("Name", "Calfem simulation of Free-Free beam."); legend; hold on; grid minor;
for mode=modes
    plot(x, psi(y_pos, mode), "DisplayName", sprintf("Mode %d", mode), "LineWidth", 1.5);
end
legend("AutoUpdate", "off")
plot(xlim, [0 0], 'k—')
for xZero=[37.864, 52.93, 89.67, 142.326, 200, 257.673, 310.329, 347.07, 362.1345]
    plot([xZero xZero], ylim, 'k—')
end
xlabel('x-[mm]'); ylabel('Mode-shape');
title('Free-free beam-Bending-modes-shapes');

disp('Free-Free-first-three-non-zero-bending-frequencies-[Hz]-(Calfem):');
fprintf("%.02f Hz\n", f_freefree_sim);

%% Task 2 Fixed-Free Sim

Lc = 300e-3;
Nel = 18;
[M, K] = calvem_beam(Nel, Lc, E0);
M = M(4:end, 4:end);
K = K(4:end, 4:end);
[psi, s] = eig(-M\K);
[nat_freq, I] = sort(imag(sqrt(diag(s)))./(2*pi));
psi_s = psi(:, I);
x = 300/Nel:300/Nel:300;
y_pos = 2:3:54;

figure("Name", "Calfem simulation of Fixed-Free beam."); legend; hold on; grid minor;
for mode=modes
    modevec = psi_s(y_pos, mode);
    plot(x, modevec-modevec(1), "DisplayName", sprintf("Mode %d", mode), "LineWidth", 1.5);
end
legend("AutoUpdate", "off")
plot(xlim, [0 0], 'k—')
for xZero=[Lc/2, 232.7, 263.63]
    plot([xZero xZero], ylim, 'k—')
end

```

```

end
xlabel('x-[mm]'); ylabel('Mode-shape');
title('Fixed-free beam-Bending-modes-shapes');

disp('Fixed-Free first three non-zero bending frequencies [Hz] (Calfem):');
fprintf("%.02f Hz \n", nat_freq(1:3))

%% Task 3 - Experiment settings
% Highest frequency : 1755 ~ 2000 Hz -> fs = 4000 Hz.

%% Task 4 - Free-Free Experimental
w = warning('off', 'MATLAB:table:ModifiedAndSavedVarnames');

fs = 4000;
data1 = readtable('Group2_Modal.csv', 'VariableNamingRule', 'modify');
f = data1.Frequency;
F_Autospectrum_Re = data1.Re_N2_RMS;
F_Autospectrum_Im = data1.Im_N2_RMS;

Coh = data1.Coherence;
accelerance_Re = data1.Re_Accelerance;
accelerance_Im = data1.Im_Accelerance;
accelerance = accelerance_Re + 1j*accelerance_Im;

[max1, ~] = max(abs(accelerance(f < 400)));
f1 = f(abs(accelerance) == max1);
[max2, ~] = max(abs(accelerance(400 < f & f < 1000)));
f2 = f(abs(accelerance) == max2);
[max3, ~] = max(abs(accelerance(1000 < f)));
f3 = f(abs(accelerance) == max3);
f_freefree_exp = [f1; f2; f3];

figure("Name", "Experimental Data - Accelerance FRF"); legend; hold on;
xlabel("Frequency [Hz]"); ylabel("Magnitude Accelerance [ms^{-2}N]");
title("Experimental Data - Accelerance FRF of Free-Free beam"); grid on;
plot(f, abs(accelerance), "DisplayName", "abs(Accelerance)");

scatter([f1 f2 f3], [max1 max2 max3], "DisplayName", "Resonant Frequencies")
disp("Free-Free first three non-zero bending frequencies [Hz] (Experiment):")
fprintf("%.02f Hz\n", [f1 f2 f3])

%% Task 5 - Cantilever Experimental
warning('off', 'MATLAB:table:ModifiedAndSavedVarnames');

points = 1:6; % accelerometer positions
cohThresh = 0.8; % quick quality flag

FRF_modal = struct([]);
f = readtable('Group2_Modal_A1F5.csv', 'VariableNamingRule', 'modify').Frequency;
Coherences = zeros(length(points), length(f));
H = zeros(length(points), length(f));
figure; hold on; legend;
for point = points
    data2 = readtable(sprintf('Group2_Modal_A%dF5.csv', point), 'VariableNamingRule', 'modify');

```

```

f      = data2.Frequency;          % Hz
Hr     = data2.Re_Accelerance; %  $Re(H)$   $(m/s^2)/N$ 
Hi     = data2.Im_Accelerance; %  $Im(H)$ 
Coh     = data2.Coherence;          %  $\gamma^2$ 

FRF_modal(point).point = point;
FRF_modal(point).f      = f;
FRF_modal(point).H      = Hr + 1j*Hi;
FRF_modal(point).Coh     = Coh;

H(point,:) = Hr+1j*Hi;
Coherences(point,:) = Coh;
fracBad = length(Coh(Coh<cohThresh))/length(Coh);
fprintf( 'Coh<-%.01f--Point-%d: -%.51f-%%\n', cohThresh, point, 100*fracBad );

semilogx(f, abs(H(point,:)), "DisplayName", sprintf("FRF %d", point));
grid on; xlabel( 'Frequency-[Hz]' ); ylabel( 'Magnitude-[(m/s^2)/N]' );
title( sprintf( 'Accelerance-FRF-magnitude, -A%dF5', point ));
end

%% Task 6 – Compare Experiment (Task 4) with Simulated (Task 1) Free-Free and update Young
scale_E = (f_freefree_sim \ f_freefree_exp)^2; %  $K = w^2 M$ 

E_new = E0 * scale_E;
[M_new, K_new] = calfem_beam(Nel, L, E_new);

[psi_new, D_new] = eig(K_new, M_new);
f_new = sort(sqrt(diag(D_new))/(2*pi));
f_new = f_new(f_new>1);
f_new = f_new(1:3);
fprintf( 'Experiment:-----%.02f--%.02f--%.02f-Hz-\n', f_freefree_exp);
fprintf( 'Calfem-Difference-with-E=--%.02f-GPa: ---%.02f--%.02f--%.02f-Hz-\n', E0*1e-9,
f_freefree_sim-f_freefree_exp );
fprintf( 'Calfem-Difference-with-E=--%.02f-GPa: ---%.02f--%.02f--%.02f-Hz-\n', E_new*1e-9, f_

%% Task 7 – Plot coherence of 1 FRF Task 5 (Cantilever Experimental), setup FRF matrix. Use
clc; close all;
point = 2;

figure( 'Name', sprintf( 'Task-5: -FRF-A%dF5', point ));
subplot(2,1,1);
plot(f, abs(H(point,:)), "DisplayName", sprintf("FRF %d", point));
grid on; xlabel( 'Frequency-[Hz]' ); ylabel( 'Magnitude-[(m/s^2)/N]' );
title( sprintf( 'Accelerance-FRF-magnitude, -A%dF5', point ));
subplot(2,1,2);
plot(f, Coherences(point,:)); ylim([0 1]); grid on;
xlabel( 'Frequency-[Hz]' ); ylabel( 'Coherence-\gamma^2' );
title( sprintf( 'Coherence-magnitude, -A%dF5', point ));

f_freefree_exp = FRF_modal(1).f; % Hz (column)
Nf = length(f_freefree_exp);
Nout = length(FRF_modal);

```

```

H_exp = zeros(Nf, Nout, 1);
CohExp = zeros(Nout, Nf);
disptime = 5;

xin = 5;

dx = Lc/6;

outno = [1 2 3 4 5 6];
inno = xin;

xout = Lc-dx*(outno-1);
yout = zeros(length(xout), 1);
zout = yout;

for p = points
    H_exp(:,p,1) = FRF_modal(p).H;
    CohExp(p,:) = FRF_modal(p).Coh';
end

flow = 5;
fhi = f_freefree_exp(end);

hN = 700;
[h,t,fs] = impresp(H_exp, f_freefree_exp);
mif = modeind3(H_exp);
mif = mif/max(mif); % normalize

Norders = 30;
poles = complex(h, fs, hN, Norders, mif, f_freefree_exp, flow, fhi);
[residues, ~] = pol2resf(H_exp, f_freefree_exp, poles, flow, fhi, disptime); % [residues, r

[fn, zeta] = poles2fd(poles); % [Hz, %]
keep = fn > 1;
fn = fn(keep);
zeta = zeta(keep);

[fn, I] = sort(fn, 'ascend');
zeta = zeta(I);
residues = residues(I, :, :);

geodef = [xout' yout zout yout (6:-1:1)' zout];

plotseq = [1:6 0];
scale = 1/10;
refpnt = 5;
ang = 0;

for mode = 1:3
    fprintf('Mode-%d: -fn=-%7.2f-Hz, -zeta=-%6.3f-%%%\n', mode, fn(mode), zeta(mode));
    modevec = (residues(mode, :));
    a = exp(-1j*angle(modevec(1)))
    modevec = modevec * exp(-1j*angle(modevec(1))); % phase align

```

```

modevec = modevec ./ max(abs(modevec)); % normalize

modeplot(mode, geodef, plotseq, modevec, scale, refpnt, ang);
title(sprintf( 'Experimental-Mode-%d-~-%.1f-Hz', mode, fn(mode)));
grid on; axis on; axis normal
xlim([0 0.35])
end

%% Task 8 - Compare the resonance frequencies from the measurements, update clamping condi
fprintf( '\nTask-8');
fprintf( 'Measured-(Task-7)-Fixed-Free:~%.02f~%.02f~%.02f-Hz\n', fn(1:3));
fprintf( 'FE-(Task-2)-Fixed-Free:~%.02f~%.02f~%.02f-Hz\n', nat_freq(1:3));
fprintf( 'Difference-(FE---Measured):~%.02f~%.02f~%.02f-Hz\n\n', nat_freq(1:3)-fn(1:3));

target = fn(1:3); % measured cantilever modes [Hz]
fe_ideal = nat_freq(1:3); % original FE (ideal fixed) [Hz]

fprintf( 'Task-8-(manual-k_theta)');
fprintf( 'Measured-(Task-7):~%.08f~%.08f~%.08f-Hz\n', target);
fprintf( 'FE-(ideal-fixed, -Task-2):~%.08f~%.08f~%.08f-Hz\n', fe_ideal);
fprintf( 'Diff-(FE---Meas):~%.08f~%.08f~%.08f-Hz\n\n', (fe_ideal - target));

[M_new, K_new] = calvem_beam(Nel, Lc, E_new);

% Base DOF indices (global, 1..3 are node 1's ux, uy, theta):
baseUx = 1; baseUy = 2; baseTh = 3;

k_xy = 1e14; % [N/m] large springs on ux, uy
k_theta = 5*1e4; % [Nm/rad] rotational spring at the clamp

f_upd = springs(M_new, K_new, k_xy, k_theta, inno, y_pos, npoles);
f_upd3 = f_upd(1:3);

fprintf( 'Using-k_theta=~%.3g-Nm/rad\n', k_theta);
fprintf( 'Updated-FE:~%.08f~%.08f~%.08f-Hz\n', f_upd3);
fprintf( 'Diff-(Updated-FE---Meas):~%.08f~%.08f~%.08f-Hz\n\n', (f_upd3 - target));

figure( 'Name', 'Task-8:~Cantilever-Frequency-Comparison'); hold on; grid on;
bar([target f_upd3 fe_ideal]);
set(gca, 'XTickLabel', {'Mode-1', 'Mode-2', 'Mode-3'});
legend( 'Measured', 'Updated-FE-(finite-clamp)', 'Original-FE-(ideal-fixed)', 'Location', 'north');
ylabel( 'Frequency-[Hz]');
title(sprintf( 'Comparison-(k_{theta}=~%.0f-kNm, -E=~%.02f-GPa)', k_theta*1e-3, E_new*1e-9));

function fn = springs(M, K, k_xy, k_theta, inno, y_pos, npoles)
    % K(1:3, 1:3) -> (ux; uy; theta)
    K(1,1) = K(1,1) + k_xy; % Translation x
    K(2,2) = K(2,2) + k_xy; % Translation y
    K(3,3) = K(3,3) + k_theta; % Rotation stiffness

    [poles, ~] = resonances(M, 0*M, K, inno, y_pos, npoles);
    [fn, ~] = poles2fd(poles);

```

```

    fn = fn(:);
    fn = fn(fn > 1);
    fn = sort(fn, 'ascend');
end

%% functions
function [M, K, m] = calvem_beam(Nelements, l, E) % Copied from Canvas, and edited.
    % Dimensions and material data
    b = 30e-3;
    h = 10e-3;
    A = b*h;
    I = b*h^3/12;
    rho = 2700;
    ml = A*rho;
    epv = [E A I ml];

    %Topology, Elements and DOFs;
    nel = Nelements;
    dl = l/nel;
    Edof = zeros(nel, 7);
    Coord = zeros(nel+1, 2);
    Dof = zeros(nel+1, 3);
    Dof(1, :) = [1:3];
    for i = 1:nel
        Edof(i, 1) = i;
        Edof(i, 2:end) = [1:6] + ((i-1)*3);
        Dof(i+1, :) = [1:3] + (i*3);
        Coord(i+1, :) = [i*dl, 0];
    end

    % Generate element matrices and assemble in global matrices
    K = zeros((nel+1)*3, (nel+1)*3);
    M = K;
    [Ex, Ey] = coordxtr(Edof, Coord, Dof, 2);

    for i = 1:nel
        [k, m, c] = beam2de(Ex(i, :), Ey(i, :), epv);
        K = assem(Edof(i, :), K, k);
        M = assem(Edof(i, :), M, m);
    end
    m = ml*l;
end

```